

Protocol

Reference standard 485 Modbus RTU protocol CRC16 check

Serial port parameters

Baud rate: 38400

Data bits: 8

Check Digit: No Parity

stop bit: 1

Initialize default slave ID: 01

control mode

1. T10 wireless remote control mode
2. Modbus speed closed-loop control mode
3. Modbus position closed-loop control mode
4. 485 joystick control mode

Register address and description

Register function	address code (decimal)	address code (hex)	Read and write function code	register value
485 joystick (vertical)	10	0x0A	0x03,0x06,0x10	Signed hexadecimal parameter Range: 800~~-800
485 joystick (horizontal)	11	0x0B	0x03,0x06,0x10	Signed hexadecimal parameter Range: 800~~-800
Left wheel target speed	12	0x0C	0x03,0x06,0x10	Signed hexadecimal parameter
Right wheel target speed	13	0x0D	0x03,0x06,0x10	Signed hexadecimal parameter
control mode	80	0x50	0x03,0x06,0x10	0: Wireless remote control mode 1: Speed closed loop mode 3. Position closed loop mode 485 joystick control mode
start-stop command	81	0x51	0x03,0x06,0x10	0: stop, parameter configuration state 1: start
Left wheel target speed	82	0x52	0x03,0x06,0x10	Signed hexadecimal parameter
Right Wheel target speed	83	0x53	0x03,0x06,0x10	Signed hexadecimal parameter

Left wheel target mileage H	84	0x54	0x03,0x06,0x10	32-bit unsigned pulse number
Left wheel target mileage L	85	0x55	0x03,0x06,0x10	
Right wheel target mileage H	86	0x56	0x03,0x06,0x10	32-bit unsigned pulse number
Right wheel target mileage L	87	0x57	0x03,0x06,0x10	
Actual speed of left wheel	54	0x36	0x03	Signed hexadecimal parameter
Actual speed of right wheel	55	0x37	0x03	Signed hexadecimal parameter
Left wheel actual mileage H	56	0x38	0x03	32-bit unsigned pulse number
Right wheel actual mileage L	57	0x39	0x03	
Actual mileage of the right wheel H	58	0x3A	0x03	32-bit unsigned pulse number
Actual mileage of the right wheel L	59	0x3B	0x03	
Displacement loop control state	60	0x3C	0x03	1. The state when the displacement loop is detected 2. 1:0x01 : running 3. 0x00: before or after running
error code output	71	0x47	0x03	1: 0x00: normal without exception
Device ID	512	0x200	0x03,0x06	1:0x01--0xFE(1-255) 0xff is the broadcast address

1. 485 interface remote control mode

- Two-dimensional joystick data input (X, Y) corresponds to register

address (0x0A, 0x0B)

2. Input range: (800---(-800)) origin (0,0)
3. The refresh rate is not more than 1 second interval (recommended 50MS)
4. If the data interaction interval exceeds 1S, the program executes the communication interruption judgment mechanism
5. Feedback real-time speed linear speed (3000---(-3000)) positive and negative sign represents direction
6. Speed feedback value (left wheel, right wheel) corresponding register (0x36, 0x37)

2. Speed closed-loop control mode

1. Speed data input (L left wheel, R right wheel) corresponds to the register address (0x0C, 0x0D)
2. Input range: (3000---(-3000)) origin (0,0)
3. The refresh rate is not more than 1 second interval (recommended 50MS)
4. If the data interaction interval exceeds 1S, the program executes the communication interruption judgment mechanism
5. Feedback real-time speed line speed 0-3000 line speed, the speed has no positive or negative distinction
6. Speed feedback value (left wheel, right wheel) corresponding register (0x36, 0x37)

Device ID range: 0x01--0xFE(1-254) 0xFF is the broadcast address

When the user forgets the ID, the slave ID of the device can be retrieved through the broadcast address

Speed closed-loop control:

Unit: line/second

Accuracy: 20 lines

Range: 600-3000 (forward is positive input, backward is negative input)
minimum speed is 600 lines/second

Configuration parameters: speed loop mode, speed value real-time input update

Write input speed frequency: 1HZ (continuous input) Stop when data is interrupted, and the speed can be adjusted during driving

Position closed-loop control:

Speed parameters are the same as speed closed loop

Displacement parameters:

unit: line

Range: 200-100000 lines

Accuracy: 100 lines

Configuration parameters: displacement loop mode, speed value, displacement value, operation start-stop mode operation after update

After the displacement loop is completed, the real-time displacement value is cleared, and the parameters need to be configured repeatedly before the vehicle control is executed. During the driving process, the displacement and speed cannot be modified, and the start-stop function can be controlled in real time.

Read input displacement frequency 1hz

Note: After the parameter is updated, after writing the start command, the start-stop command key needs to be read at a reading frequency of 1hz. It is used to support the heartbeat command of the displacement loop movement. If it exceeds, the stop will be cleared.